

DFE Loudspeaker Crossovers

Analog Crossover Filters with Fast Roll-Off Behavior

By
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Derived from Elliptic (DFE) crossovers offer improved characteristics compared to existing filter types and can be implemented via passive, analog active, and IIR digital means. The combination of a relatively flat sum, a narrow and steep transition band, and well-controlled group delay position them as higher-performance alternatives to traditional loudspeaker crossovers.

The crossover system is an essential component of all multi-way loudspeakers. Butterworth and Linkwitz-Riley filters are by far the most commonly used types, with a few others making up the rare exception. Despite nearly 100 years of loudspeakers with crossovers, the variety in response characteristics is surprisingly small. The ultimate roll-off rate, or steepness, of all-pole filters is fixed at 6dB per pole and response differences are primarily found near the passband edge where the response enters the transition band.

Because the response of the filter in the time domain is determined by the frequency domain response, here too the differences are not all that significant. Although finite impulse response (FIR) digital filtering removes these limitations and opens a vast array of choices to the designer, FIR filters are computationally demanding, cannot be realized in the analog domain, and give rise to latency during audio playback.

Steep loudspeaker crossover filters offer several advantages: they reduce the need to keep the relative phase angle between drivers well matched over a wide range of frequencies; with less frequency overlap between adjacent drivers the separation distance becomes less critical; shifts in the main lobe in the loudspeaker's radiation pattern and other unwelcome off-axis effects are reduced [1]. Unfortunately, the steepness of high-order all-pole filters is directly related to their group delay and high group delay is attributed to poor sound quality. High-order all-pole filters are also more complex to implement. These factors give rise to limits on the amount of steepness that is practical with all-pole crossovers.

I have developed a new family of loudspeaker crossover filters by modifying textbook elliptic filters to improve their suitability as loudspeaker crossovers. These "derived from elliptic" (DFE) filters

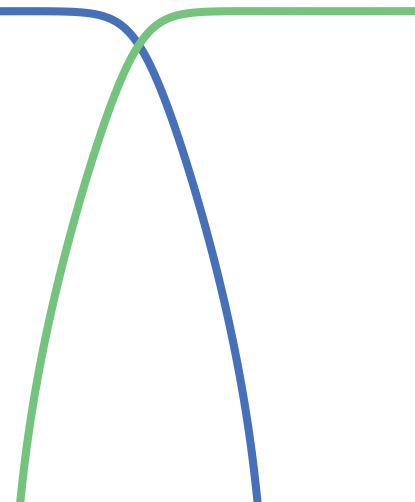
offer improved characteristics compared to other analog domain filters typically used in loudspeaker crossovers. The high- and low-pass responses have a relatively flat sum, a narrow transition band with steep first-octave rolloff, and controlled group delay yet can be implemented in the analog domain. The different response shapes and improved performance characteristics of DFE crossovers provide increased freedom and flexibility when choosing how to integrate the drivers of a loudspeaker system.

Loudspeaker Crossover Filter Requirements

The requirements for loudspeaker crossover filters are slightly different from other filtering tasks in electronics because a pair of low-pass and high-pass responses must work together to provide several desirable characteristics. In his book, *The Design of Active Crossovers* [2], Douglas Self enumerates these as:

1. Adequate flatness of summed response
2. Sufficiently Steep Roll-off
3. Acceptable Polar Response
4. Acceptable Phase Response
5. Acceptable Group Delay Behavior

These encompass the response in both the frequency and time domains: items 1 and 2 describe the frequency domain behavior, items 4 and 5 the time domain behavior, and item 3 is simply a consequence of item 4. It's not generally possible to simultaneously optimize most or all these characteristics unless FIR digital filtering is used. That type of filtering will not be covered in this article. Instead, we will focus on real-time analog or IIR digital filtering applications.



Because DFE crossovers are not all-pole, and their poles are not all positioned at the same frequency, they have a different response shape and offer a superior combination of frequency and time domain characteristics. DFE filters maximize the steepness in the transition band (item 2) by allowing the group delay to rise towards the limit of audibility (item 5). Another way to describe the limit of audibility of (excessive) group delay is as the threshold where phase distortion becomes audible. To better familiarize the reader with group delay, a brief review of the topic will now be given. The phase response of a filter is related to when the energy of an impulse applied to the filter input appears at its output and the group delay is simply a different way to express this relationship. When a filter has a linear phase characteristic its group delay will be constant (sometimes referred to as “flat” group delay) because group delay is proportional to the first derivative of the phase response (e.g., $\tau = -\partial\phi/\partial\omega$).

The amplitude and group delay responses of Bessel, Butterworth, and Elliptic low-pass filters are compared in **Figure 1**. These plots illustrate how each filter type trades off frequency domain steepness for group delay flatness. For filters having a 90-degree relative phase angle like the odd-order Butterworth types, one choice for the polarity will result in a much higher level of group delay than the other but the on-axis amplitude response remains unaffected.

The group delay of a crossover is a combination of the group delay responses of its low-pass and high-pass filters weighted by their amplitude response and may be characterized by the following trends: a finite amount of group delay is present at DC, the group delay decays to zero at high frequency, and some other amount of group delay is observed at the crossover point (F_c) that is related to how quickly the response is changing between the passband and the transition band. When the group delay near F_c is elevated above the level at DC, this is called group delay “peaking.” When there is strong peaking in the group delay near F_c , the time domain signal will often display ringing at this frequency. This in itself can be audible, or it can change the perception of signals with sharp transients such as isolated click sounds when heard through headphones.

A recent publication by Juho Liski, Aki Mäkitvirta, and Vesa Välimäki in *The IEEE/ACM Transactions on Audio Speech and Language Processing* [3] reviewed the available literature regarding the audibility limit of group delay as function of frequency for a variety of signals. For music signals, the group

delay audibility limit lies between 1.5ms and 2ms at 1kHz, remaining relatively unchanged at higher frequencies. Below 1kHz the audibility limit increases, approximately following a $1/f$ relationship. The data reviewed in reference 3 provides clarity on this important aspect of human hearing and restricts the type of filters that are suitable for high performance loudspeaker crossovers.

The plots in Figure 1 show that the elliptic filter has very high group delay peaking at F_c compared to the Butterworth type, greatly exceeding the audibility limit. This is primarily because these filters place a high Q pole close to F_c , and high Q poles give rise to group delay peaking. This pole, however, is necessary for supporting the frequency response at the edge of the passband where it enters the

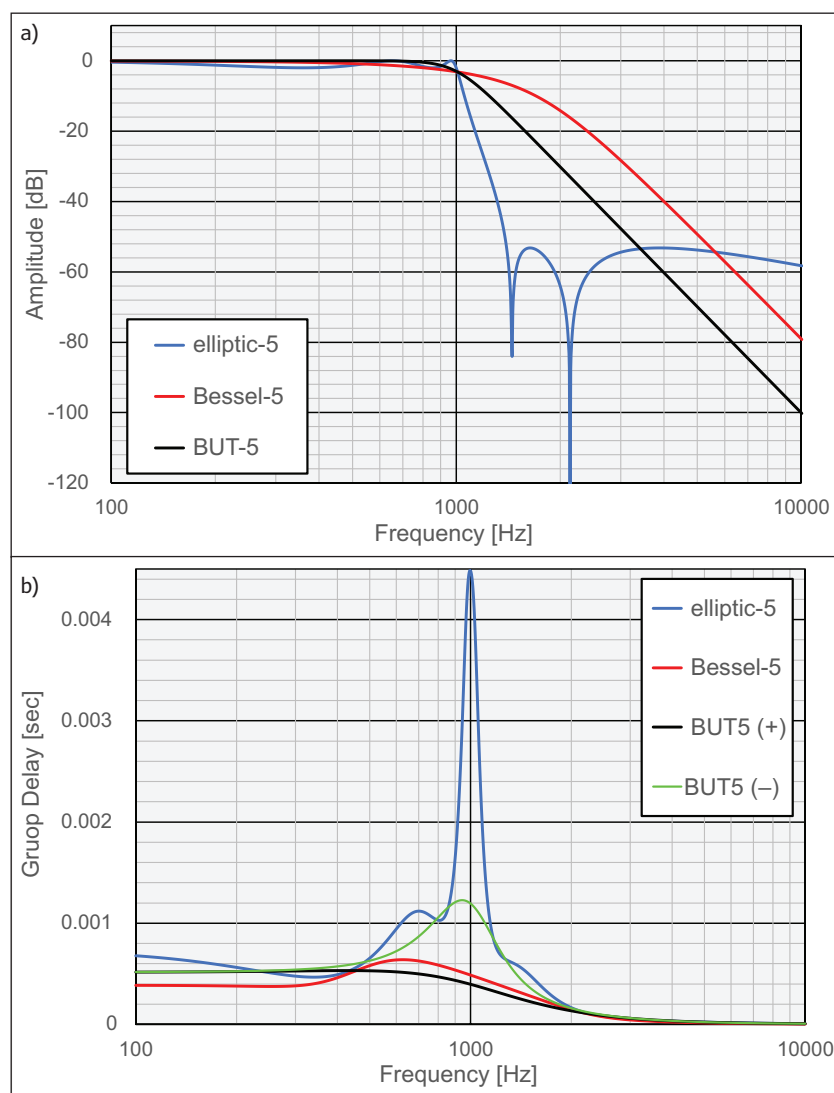
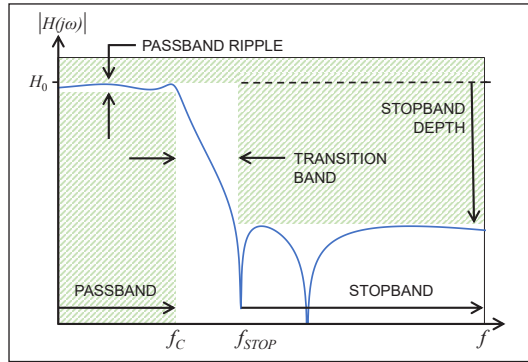


Figure 1: Low-pass amplitude responses (a) and crossover group delay responses (b) for fifth-order elliptic, Bessel, and Butterworth crossovers. The group delay for both the normal and reversed polarity versions of the Butterworth crossover are shown.

Figure 2: An illustration of, and terms associated with, filter specification. H_0 is the nominal passband level, typically 0dB.



transition band. Despite the group delay behavior, the transition band steepness of elliptic filters is unsurpassed and is why they are used whenever a very abrupt frequency cutoff is needed (e.g., for anti-aliasing and similar band-limiting tasks).

If we put aside the problem of group delay peaking for a moment, we can push forward and try to form a crossover by mirroring a low-pass elliptic filter about F_c to create the complimentary high-pass filter and then compute the sum of these two filters. Unfortunately, the result is almost always wholly unsatisfactory. At F_c there is either sharp peaking, a dip if the phase of one filter is reversed, or other types of frequency response problems. It would seem that these behaviors violate most of the desirable characteristics for crossovers outlined earlier. Yet, if these filters could be manipulated in such a way as to flatten the passband sum and reduce group delay peaking, they would be very useful as loudspeaker crossovers. My work in this area has led to the DFE crossover family. The DFE crossovers solve the problems presented by elliptical filters while retaining the steep roll-off and narrow transition band that makes elliptic filters attractive for loudspeaker crossover applications. At a glance, many DFE crossover filters resemble, but are not the same as, Chebyshev Type II (inverse Chebyshev) filters.

DFE Filter Synthesis

DFE crossover filters are based on elliptic filters, which are broader and more varied than the all-pole filters. For a given order, an elliptic filter can trade off passband ripple amplitude, stopband depth, and the stopband edge frequency within a range of values. Often there are multiple combinations of these parameters that produce successful but slightly different DFE crossovers. Increasing the filter order and/or the location of the first stopband zero influences both the depth of the stopband and the width (narrowness) of the transition band. The reader may refer to **Figure 2** to better understand the concepts of passband ripple amplitude, stopband, stopband depth, and transition band.

The process of creating a DFE crossover from an elliptic filter is guided by a fitness metric that reflects the desirable features of loudspeaker crossovers such that the lower the value of the fitness metric the better the fitness. The metric includes characteristics such as the relative phase angle between the high-pass (HP) and low-pass (LP) filter responses, the deviation of the LP+HP sum from "flat," the group delay of the LP+HP sum, and the deviation of the LP filter response from the filter specifications (Figure 2). The process begins with an elliptic filter candidate of order $N > 2$, written as a series cascade of first- and second-order filter blocks. Each block has a pole frequency (F_p) and second-order blocks also have a pole quality factor (Q) and zero frequency (F_z). For an N th order filter this leads to either $3*N$ parameters (when the filter order is even) or $3*(N-1)+1$ parameters (when the filter order is odd). True elliptic filters are comprised only of second-order low-pass-notch or high-pass-notch filters, with odd-order elliptic filters also including a single first-order high-pass or low-pass filter stage. For a review of high-pass and low-pass notch filters, see my article in the August issue of *audioXpress* [4].

To generate a DFE crossover candidate, a nonlinear optimization engine manipulates the values of the $3*N$ or $3*(N-1)+1$ parameters in an attempt to minimize the value of the fitness metric. The result may or may not be satisfactory and the process may be repeated from a different starting point. Sometimes, the polarity of the high-pass filter is reversed and a new attempt at optimization is performed. The challenge lies in the complexity of the elliptic response, the presence of irregular features and local minima in the error surface, and the fact that the fitness metric inherently contains measures of the frequency domain and time domain behavior of the filter that are not independent of one another.

About the Author

Charlie Laub holds a bachelor's degree from the University of Massachusetts and a Ph.D. from the University of Connecticut, both in Chemical Engineering. He has been building and learning about loudspeakers since he was 9 years old, and in his retirement enjoys applying his analytical skills and love of music to DIY audio. He has been experimenting with and building dipole and open baffle loudspeakers for more than 10 years and has developed several public-domain software tools related to acoustic measurement processing and crossover design such as the FRD Blender and Minimum Phase Extractor, and the ACD Crossover Design suite. He is a strong proponent of DSP processing in software on lightweight Linux platforms such as the Raspberry Pi and other mini PCs, and he has written several audio processing plug-ins for this purpose. At present, Charlie and his wife Birgit live in Bavaria, Germany, having recently relocated from Michigan.



[ORDER][TYPE]_[STOPBAND_FREQUENCY]_A[STOPBAND_DEPTH]_P[PHASE_ANGLE_AT_FC]

ORDER is the order of the high-pass and low-pass filters used to construct the crossover.

TYPE can consist of Derived from Elliptic (DFE), DFE Squared (DFES), and All-Pole plus DFE (AP-DFE).

STOPBAND_FREQUENCY is the frequency at which the stopband region begins, normalized by F_c , for the low-pass filter.

STOPBAND_DEPTH is the attenuation below which the response remains in the stopband.

PHASE_ANGLE_AT_FC is the relative phase angle between low-pass and high-pass crossover filters at the crossover frequency.

Table 1: An explanation of the structure of DFE filter nomenclature, which conveys information about the filter performance.

Examples of DFE Crossovers

Since there are multiple DFE crossovers for a given order, a nomenclature is employed that also includes information about the amplitude and relative phase angle at F_c to more easily distinguish between them. The form of this nomenclature is described in **Table 1**. The following information is also provided for each crossover:

Group Delay: To a first-order approximation, the group delay behavior of a DFE crossover can be described by the group delay at two frequencies: at DC and near F_c . Therefore, these values are listed for a crossover frequency of 1kHz so that comparison to the reported limits of audibility [3] is easy. A plot of the group delay versus frequency for each crossover can be found in the online supplementary material (see Project Files).

Frequency Response: Only the low-pass response for a selection of DFE crossovers is presented below, with the high-pass response being its reflection about F_c .

Ripple Amplitude: The amplitude of the sum of lowpass and highpass filters is reported as the maximum deviation X from the mean response level, resulting in a ripple band of $\pm X$ dB.

Butterworth Order for Attenuation Equivalence: The order required for a Butterworth filter to reach the stopband depth A at F_{stop} may be obtained from the relationship $A = 10 \cdot \log_{10} (1 + (F_{stop}/F_c)^{2n})$ [5]. This is provided in the description as a comparison to the order of the DFE crossover.

Examples of Fifth-Order DFE Crossovers: The 5DFE_1.52_A39_P70 (**Figure 3**) reaches a stopband depth of 39dB at $1.52 \cdot F_c$, has a phase angle of 70 degrees at F_c , a peak group delay of 0.57msec, 0.41msec at DC, and a maximum deviation within the passband of 0.06dB. A Butterworth filter of order 11 would be required to reach 39dB attenuation at $1.52 \cdot F_c$.

Another fifth-order DFE crossover, the 5DFE_1.54_A49_P5 (**Figure 4**), has a stopband

that reaches 10dB lower to 49dB despite a similar stopband edge frequency of $1.54 \cdot F_c$. Interestingly, the relative phase angle at F_c only differs from that of the 5DFE_1.52_A39_P70 by 65 degrees.

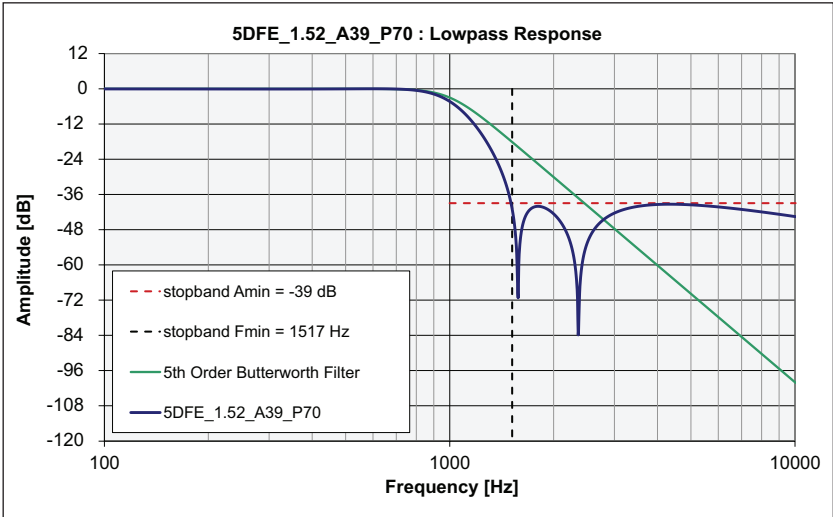


Figure 3: The 5DFE_1.52_A39_P70 low-pass response compared to a Butterworth filter of the same order.

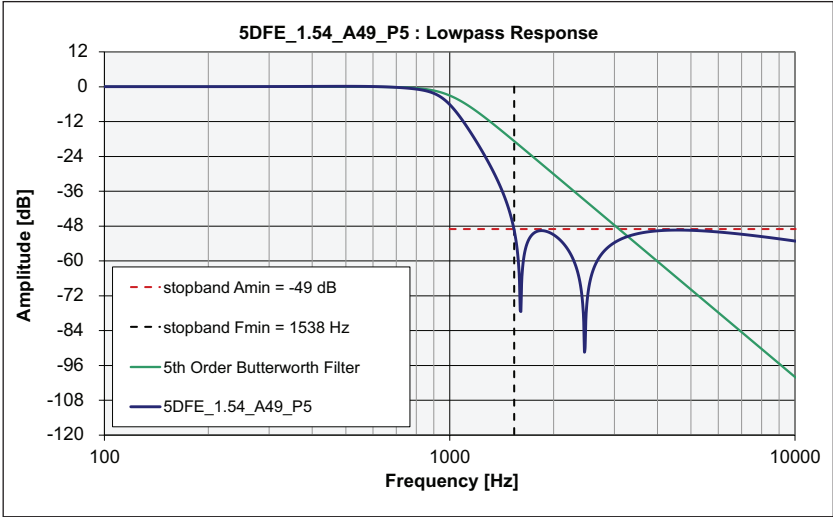


Figure 4: The 5DFE_1.54_A49_P5 low-pass response compared to a Butterworth filter of the same order.

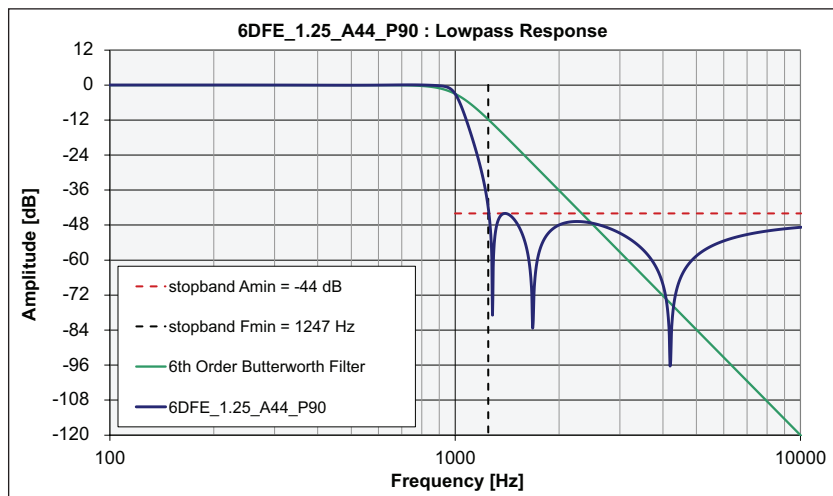


Figure 5: The 6DFE_1.25_A44_P90 low-pass response compared to a Butterworth filter of the same order.

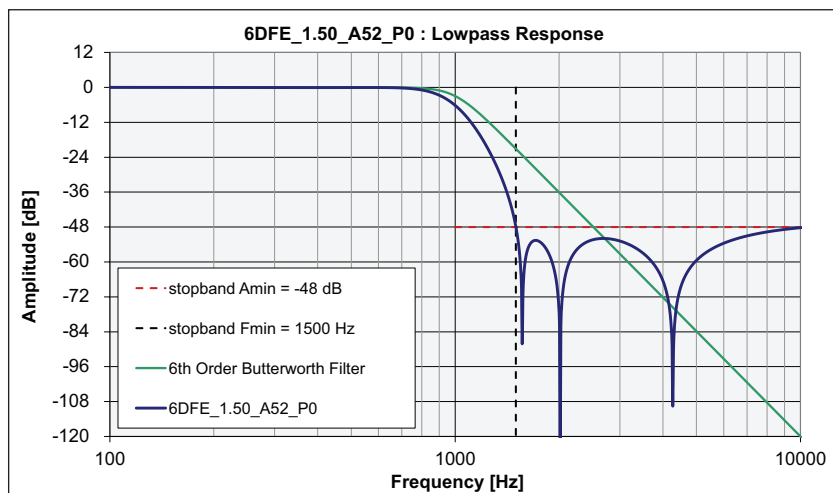


Figure 6: The 6DFE_1.50_A52_P0 low-pass response compared to a Butterworth filter of the same order.

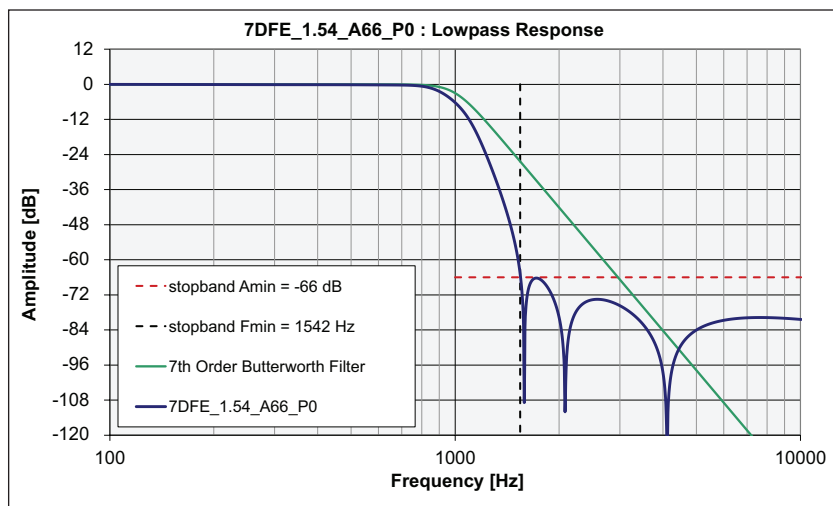


Figure 7: The 7DFE_1.54_A66_P0 low-pass response compared to a Butterworth filter of the same order.

This is a consequence of the different pole and zero frequencies and Q values used to create each crossover and that the polarity of the high-pass filter section of 5DFE_1.54_A49_P5 is reversed. 5DFE_1.54_A49_P5 has a peak group delay of 1.19msec, 0.52msec at DC, and a maximum deviation within the passband of 0.20dB. A Butterworth filter of order 13 would be required to reach 49dB attenuation at $1.54 \cdot F_c$.

Examples of Sixth-Order DFE Crossovers: The 6DFE_1.25_A44_P90 (Figure 5) is a remarkable example of the capabilities of the DFE crossover family. This filter provides 44dB of attenuation at only 1.25 times F_c . This is equivalent to an average roll-off rate of 136dB/octave over this interval, yet the normalized group delay remains relatively low. The filter is a constant power type with a phase angle of 90 degrees at F_c , a peak group delay of 1.10msec, 0.53msec at DC, and a maximum deviation within the passband of 0.08dB. A Butterworth filter of order 23 would be required to reach 44dB attenuation at $1.25 \cdot F_c$.

The 6DFE_1.50_A52_P0 (Figure 6) moves the stopband edge to $1.5 \cdot F_c$, allowing a deeper stopband of 48dB to 52dB. The LP and HP filters are in-phase at F_c . The crossover has a group delay peak of 1.16msec, 0.53msec at DC, and a maximum deviation within the passband of 0.12dB. A Butterworth filter of order 15 would be required to reach 52dB attenuation at $1.50 \cdot F_c$.

Examples of Seventh- and Eighth-Order DFE Crossovers: The seventh-order 7DFE_1.54_A66_P0 (Figure 7) has a stopband depth of 66dB that is reached at $1.58 \cdot F_c$ and deepens past 73dB above 2.5kHz, is in-phase at F_c , a maximum passband amplitude deviation 0.12dB, but a somewhat elevated peak group delay of 1.59msec, 0.73msec at DC. This is an example of how DFE crossovers do not follow the usual rules because the order is odd but high- and low-pass responses are in-phase at F_c . A Butterworth filter of order 18 would be required to reach 66dB attenuation at $1.54 \cdot F_c$.

Much like 6DFE_1.25_A44_P90, the 8DFE_1.38_A71_P84 (Figure 8) has extremely steep roll-off and provides a stopband depth of 71dB. The stopband is reached at only $1.38 \cdot F_c$, corresponding to an average rate of 153dB/octave over this interval. The response has a constant-power-like phase angle at 84 degrees, a peak group delay of 1.50msec, 0.73msec at DC, and a maximum deviation within the passband of only 0.03dB. A Butterworth filter of order 26 would be required to reach 71dB attenuation at $1.38 \cdot F_c$.

DFE Squared Crossovers: The DFE crossover design process can be expanded to other sorts of filters that are useful for loudspeaker crossovers. Consider Linkwitz-Riley crossovers—these are formed by placing two identical Butterworth filters in series and are sometimes called “Butterworth Squared” filters. Optimizing two low-order DFE crossovers in series gives rise to DFE-Squared or DFES crossovers. Examples of DFES crossovers can be found in the online supplementary material.

Combining All-Pole and DFE Crossovers: Another interesting variation can be obtained by combining an Nth order DFE crossover with an Mth order all-pole crossover. This combination is especially well-suited to hybrid passive+active crossover systems. There are several examples of this for the case of $N=2$ and various values of $M \geq 2$ that were described by Neville Thiele in his NTM crossover patent [6]. Very similar responses can be obtained using the DFE crossover development technique. Here the pole frequency and Q of the all-pole filters also undergo optimization. The nomenclature is extended by prepending AP and the order of the all-pole filter to the filter name.

Examples of AP+DFE Crossovers: The response of the AP2_2DFE_1.82_A38_P15 (Figure 9) is nearly identical to the fourth-order crossover described by Thiele when his factor k is set equal to 0.500. This has a stopband edge at $1.82 \cdot F_c$, stopband depth of at least 38dB with attenuation further increasing at 12dB/octave for $F > 2.75 \cdot F_c$, a phase angle of 15 degrees at F_c , a peak group delay of 0.56msec, 0.41msec at DC, and a passband amplitude deviation of 0.11dB. A Butterworth filter of order 8 would be required to reach 38dB attenuation at $1.82 \cdot F_c$.

The AP3_4DFE_1.41_A46_P83 (Figure 10) combines third-order all-pole and fourth-order DFE filters. The stopband is reached at $1.41 \cdot F_c$ and is a minimum of 46dB, falling at 18dB/octave for $F > 2.9 \cdot F_c$. The crossover has a relative phase angle at F_c of 83 degrees, a peak group delay of 0.93msec, 0.61msec at DC, and a maximum passband amplitude deviation of less than 0.01dB. A Butterworth filter of order 16 would be required to reach 46dB attenuation at $1.41 \cdot F_c$.

Discussion

All the DFE crossovers described here were designed with a crossover frequency (F_c) equal to 1kHz in order to make it easy to judge their group delay (GD) and group delay peaking (GDP) against the threshold of audibility (1.5msec to 2msec at 1kHz) [3]. The audibility threshold

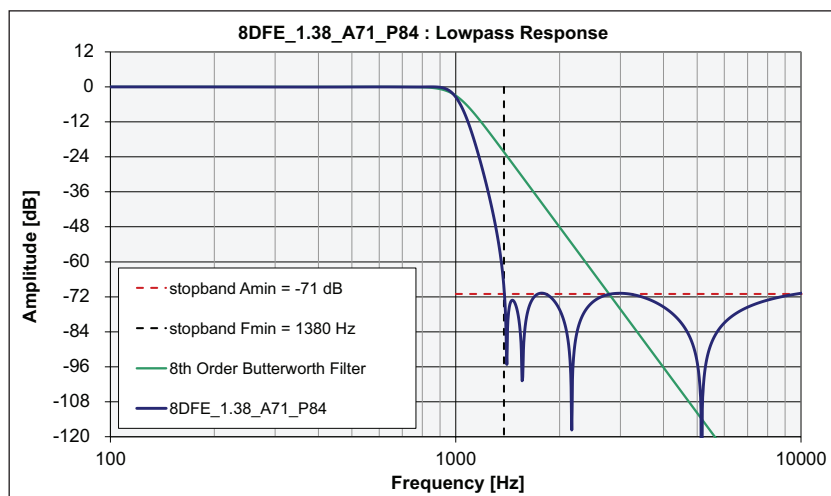


Figure 8: The 8DFE_1.38_A71_P84 low-pass response compared to a Butterworth filter of the same order.

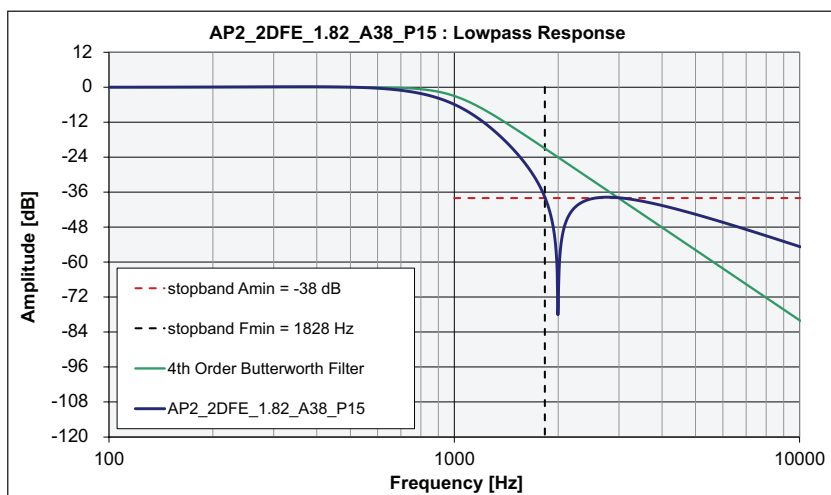


Figure 9: The AP2_2DFE_1.82_A38_P15 low-pass response compared to a Butterworth filter of the same order.

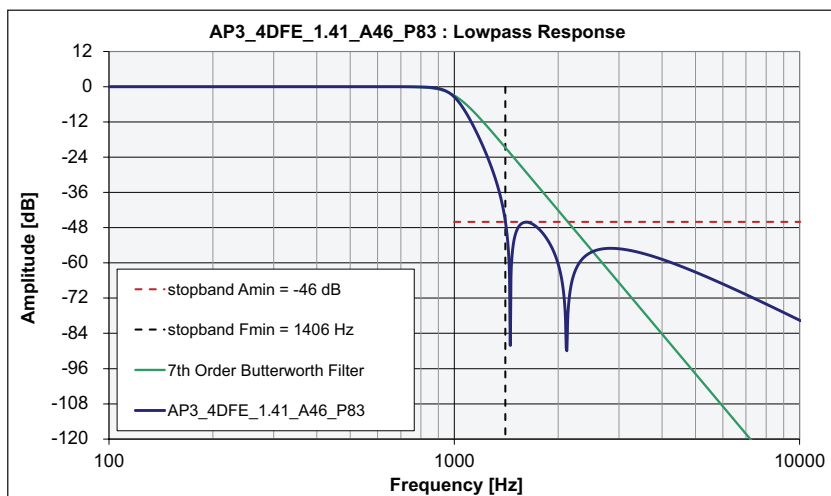


Figure 10: The AP3_4DFE_1.41_A46_P83 low-pass response compared to a Butterworth filter of the same order.

varies with frequency roughly following a type of “hockey stick” trend: Above 1kHz the limit of audibility remains approximately the same, and below 1kHz the audibility limit is increasing following a $1/f$ relationship. On the other hand, the group delay and peaking for crossovers follows the $1/f$ relationship for all frequencies. This means that when a crossover’s GD falls below the threshold at 1kHz it will be below it for all other frequencies as well. An additional consequence is that, for F_c above 1kHz, even if the amount of GD at 1kHz is relatively high, GD quickly becomes less than the audibility limit when F_c is increased to a higher frequency.

Most of the DFE filters described in this article require a Butterworth filter of order 11 to 23 to achieve equivalent attenuation—none of these would be acceptable due to their high GD and GDP [3]. Typically, the order for a Butterworth filter will be at least 2 but often 3 times or more than a DFE filter to provide equivalent “first octave attenuation” and for loudspeaker crossovers it is this portion of the transition band that is most important for band definition. The un-beatable first octave rate of attenuation is one reason why DFE filters are a new and interesting option for loudspeaker crossovers.

Useful DFE loudspeaker crossovers have a relatively narrow transition band and are of order 4 to 8. Implied in desirable crossover characteristic (item 2) is that there must be a sufficient amount of attenuation reached at some frequency in the stopband. If the attenuation is less than about 40dB the out-of-band signal is not sufficiently suppressed, and only DFE crossovers of at least order 4 reach a stopband attenuation of 35dB to 40dB while still being relatively steep. A different problem occurs

when the DFE crossover order is too high—the GDP cannot be made sufficiently low to fall below the audible threshold [3], and this violates desirable characteristic (item 5). When a DFE crossover is designed with a wide transition band it will have a roll-off characteristic similar to a Butterworth of the same order for the first octave or so, which does not offer anything new or interesting. Therefore, there is a “sweet spot” of order and steepness within which all DFE crossovers of merit are found.

DFE crossovers are “analog” in the sense that they can be realized by analog circuits and are primarily intended for line-level applications. As described in my August 2024 *audioXpress* article [4], the high-pass and low-pass notch filters required to implement DFE crossovers can be easily implemented via active analog or IIR digital filters (the DSP equivalent of an analog active filter).

A fully passive realization via LC ladder filters is also possible, however, the design and tuning is more complex and complicated due to the interaction of components and the variation in the values of real-world passive components. The design of passive ladder filters used in DFE crossovers typically requires a fixed source impedance and identical load impedance, and this makes speaker level implementation impractical.

Summary

This article has provided a small sample of the DFE crossovers that have been developed to date, and still others may be forthcoming. Detailed information on more than 30 different DFE crossovers can be found in the online supplementary material. Each DFE crossover is described via tables of its first- and second-order building blocks. Combined with the information provided in reference 4, readers will be able to implement these crossover filters in their own projects. The steep roll-off behavior of the DFE crossover filters cannot be achieved at a similar order using any other filter type, and the many ways that the DFE crossover family breaks from traditional behavior opens new and interesting possibilities for high performance crossover design. 

Project Files

To download additional materials for this article, visit <http://audioxpress.com/page/audioXpress-Supplementary-Material.html>.

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